
Online Examination Report

Date:	15.08.2014
Attempts:	1
Examinee:	Ekwee, Simon
Date of birth:	
Subject:	020-AGK
Result:	34,99 %
Time used:	00:01:00 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	2759	21.	AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	1,00	1,00
2	3648	21.	AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	0,00	1,00
3	2401	21.2.3	Aeroplane: Wings, tail and control surfaces	0,00	1,00
4	690	21.2.3	Aeroplane: Wings, tail and control surfaces	0,00	1,00
5	2398	21.2.3	Aeroplane: Wings, tail and control surfaces	0,00	1,00
6	1503	21.2.5	Helicopter : Flight controls structural aspects	0,00	1,00
7	2358	21.3.1	Hydro-mechanics: basic principles	0,00	1,00
8	158	21.6	PNEUMATICS - PRESSU AND AIRCO SYSTEMS	1,00	1,00
9	448	21.6.3	Aeroplane: Pressurisation and air conditioning system	0,00	1,00
10	2357	21.6.3	Aeroplane: Pressurisation and air conditioning system	1,00	1,00
11	2002	21.8.2	Turbine engine	0,00	1,00
12	817	21.9.3.1	DC Generation	1,00	1,00
13	251	21.9.3.2	AC Generation	0,00	1,00
14	1286	21.9.3.2	AC Generation	1,00	1,00
15	216	21.10.1.2	Engine: design, operation etc	0,00	1,00
16	2430	21.10.10.1	Performance	0,00	1,00
17	134	21.10.10.1	Performance	1,00	1,00
18	1001	21.12.2.2	Fire detection	1,00	1,00
19	312	21.13	OXYGEN SYSTEMS	0,00	1,00
20	584	21.13	OXYGEN SYSTEMS	0,00	1,00
21	3097	22.	INSTRUMENTATION	0,00	1,00
22	2824	22.	INSTRUMENTATION	1,00	1,00
23	4026	22.	INSTRUMENTATION	0,00	1,00
24	3102	22.	INSTRUMENTATION	1,00	1,00
25	3087	22.	INSTRUMENTATION	1,00	1,00
26	2822	22.	INSTRUMENTATION	0,00	1,00
27	1715	22.1.3	Fuel gauge	0,00	1,00
28	119	22.2.1	Pressure measurement	1,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	1045	22.2.4	Altimeter	0,00	1,00
30	694	22.2.4	Altimeter	0,00	1,00
31	1963	22.2.6	Airspeed indicator (ASI)	0,00	1,00
32	87	22.2.6	Airspeed indicator (ASI)	0,00	1,00
33	1158	22.2.6	Airspeed indicator (ASI)	1,00	1,00
34	1198	22.2.7	Mach meter	0,00	1,00
35	1374	22.2.7	Mach meter	0,00	1,00
36	1810	22.3	MAGNETISM - DIRECT READING COMPASS AND FLUX VALVE	0,00	1,00
37	1800	22.3	MAGNETISM - DIRECT READING COMPASS AND FLUX VALVE	1,00	1,00
38	1514	22.4.2	Rate of turn and slip indicator/Turn Co-ordinator	0,00	1,00
39	1711	22.4.2	Rate of turn and slip indicator/Turn Co-ordinator	0,00	1,00
40	884	22.6.3	Flight Director : design and operation.	1,00	1,00
Total:35%				14,00	40,00

Attachment 2: List of result by topic

Licence

Date: 15.08.2014

Examinee: Ekwee, Simon

Location: Uganda Civil Aviation Authority

Your Result: 14,00 from 40,00 Points (35,00 %)

Topics	Ammount of Questions	Points achieved	Maximum Points	Percent
21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	20	7,00	20,00	35,00
21.2. AIRFRAME	4	0,00	4,00	0,00
21.3. HYDRAULICS	1	0,00	1,00	0,00
21.6. PNEUMATICS - PRESSU AND AIRCO SYSTEMS	3	2,00	3,00	66,67
21.8. FUEL SYSTEM	1	0,00	1,00	0,00
21.9. ELECTRICS	3	2,00	3,00	66,67
21.10. PISTON ENGINES	3	1,00	3,00	33,33
21.12. PROTECTION AND DETECTION SYSTEMS	1	1,00	1,00	100,00
21.13. OXYGEN SYSTEMS	2	0,00	2,00	0,00
22. INSTRUMENTATION	20	7,00	20,00	35,00
22.1. SENSORS AND INSTRUMENTS	1	0,00	1,00	0,00
22.2. MEASUREMENT OF AIR DATA PARAMETERS	8	2,00	8,00	25,00
22.3. MAGNETISM - DIRECT READING COMPASS AND FLUX VALVE	2	1,00	2,00	50,00
22.4. GYROSCOPIC INSTRUMENTS	2	0,00	2,00	0,00
22.6. AEROPLANE : AUTOMATIC FLIGHT CONTROL SYSTEMS	1	1,00	1,00	100,00
Total	40	14,00	40,00	35,00

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1

The first indication of carburetor ice in an aircraft with a fixed-pitch propeller is

☐ an increase in manifold pressure.

☐ a decrease in manifold pressure.

☒ a decrease in RPM.

2

2-cycle engine thrust and fuel efficiency can be greatly compromised when

- ☐ exhaust systems are installed that are not specifically tuned for an engine.
- ☒ intake valve lifters fail to pressurize and provide adequate fuel to the combustion chamber.
- ☐ carbon deposits build up on exhaust valves.

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3

Which is a purpose of leading-edge flaps?

- ☒ Reduce lift without increasing airspeed.
- ☐ Increase the camber of the wing.
- ☐ Direct airflow over the top of the wing at high angles of attack.

4

The advantage of mounting the tailplane on top of the vertical stabilizer is :

- ☐ that it does not require a de-icing system
- ☐ to withdraw it from the influence of wing turbulence
- ☒ to have greater effectiveness at high speed
- ☐ to decrease fuel consumption by creating a tail heavy situation

5

What is the purpose of a control tab?

- ☒ Reduce control forces by deflecting in the proper direction to move a primary flight control.
- ☐ Move the flight controls in the event of manual reversion.
- ☐ Prevent a control surface from moving to a full-deflection position due to aerodynamic forces.

6

The mixture control for a carburettor achieves its control by:

- ☐ varying the fuel supply to the main discharge tube.
- ☒ varying the air supply to the main discharge tube.
- ☐ altering the depression on the main discharge tube.
- ☐ moving the butterfly valve through a separate linkage to the main throttle control.

7

In a hydraulic system, the reservoir is pressurized in order to:

- ☒ keep the hydraulic fluid at optimum temperature
- ☐ prevent pump cavitation
- ☐ seal the system
- ☐ reduce fluid combustibility

8

A modern Auxiliary Power Unit (APU) is designed to provide power for ground starting of an engine. It also supplies both, in the air (subject to certification limitations) and on the ground :

- ☐ air conditioning and thrust in the event of engine failure.
- ☐ either air conditioning or electrical services, but never both at the same time.
- ☒ air conditioning and electrical services.
- ☐ air conditioning and electrical services (on the ground) electrical and hydraulic back-up services (in the air).

9

The term "pressure cabin" applies when an aeroplane :

- ☐ has the ability to maintain a constant cabin altitude at all flight altitudes.
- ☐ is only pressurised in the area of the control cabin.
- ☒ has the means to maintain cabin pressure higher than ambient pressure.
- ☒ has the ability to maintain a constant cabin differential pressure at all flight altitudes.

10

In a bootstrap system, the purpose of the heat exchangers is to:

- ☐ allow a steady compressor outlet temperature.
- ☒ cool bleed air and compressor air from the turbo refrigerator.
- ☐ cool bleed air.
- ☐ allow a homogeneous temperature by mixing air flows from various air conditioning groups in operation.

11

In the stator of a turbine, the speed V and static pressure P_s vary as follows:

☐ V decreases, P_s decreases.

☐ V decreases, P_s increases.

☒ V increases, P_s increases.

☐ V increases, P_s decreases.

12

The reason for using inverters in an electrical system is

- ☒ To change DC into AC.
- ☐ To change AC into DC.
- ☐ To change the DC voltage.
- ☐ To avoid a short circuit.

13

The frequency of an AC generator is dependent on the :

- ☐ field strength and the speed of the moving part.
- ☒ number of individual poles and the field strength.
- ☐ number of pairs of poles and the speed of the moving part.
- ☐ number of individual poles only.

14

A unit that converts electrical DC into AC is :

☐ a transformer rectifier unit.

☐ a thermistor.

☐ an AC generator.

☒ an inverter.

15

Apart from flight into known icing conditions, the intake system of a diesel engine:

- ☐ *is never heated because a diesel engine is a injection engine.*
- ☒ *needs to be heated at low power settings, due to vaporisation heat.*
- ☐ *is never heated because diesel engines operate at very high temperatures.*
- ☐ *needs to be heated to prevent ice on the air filter.*

16

What criteria determines which engine is the 'critical' engine of a twin-engine airplane?

- ☐ The one with the center of thrust closest to the centerline of the fuselage.
- ☒ The one with the center of thrust farthest from the centerline of the fuselage.
- ☐ The one designated by the manufacturer which develops most usable thrust.

17

The kind of compressor normally used as a turbocharger is:

☐ *a hybrid compressor.*

☐ *an axial compressor.*

☒ *a radial compressor.*

☐ *a piston compressor.*

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18

Smoke detectors fitted on transport aircraft are of the following type :

☐ chemical

☐ electrical

☒ optical or ionization

☐ magnetic

19

When selected to normal, the oxygen proportion of the air/oxygen mixture supplied by the cockpit oxygen system regulator:

- ☒ **is 100 %.**
- ☐ **increases when the altitude increases.**
- ☐ **is constant whatever the altitude.**
- ☐ **decreases when the altitude increases.**

20

A diluter demand oxygen regulator :

- ☐ delivers oxygen flow when inhaling.
- ☒ is only recommended for use with smoke in the cockpit.
- ☐ delivers oxygen flow only above FL 100.
- ☐ mixes air and oxygen in a passenger oxygen mask.

21

How can a pilot identify a lighted heliport at night?

- ☐ White and red beacon light with dual flash of the white.
- ☐ Green, yellow, and white beacon light.
- ☒ Green and white beacon light with dual flash of the white.

22

Prior to starting the engine, the manifold pressure gauge usually indicates approximately 29" Hg. This is because the

- ☐ throttle is closed, trapping high air pressure in the manifold.
- ☒ pressure within the manifold is the same as atmospheric pressure.
- ☐ pointer on the gauge is stuck at the full-power indication.

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23

Which is the maximum tolerance for the VOR indication when the CDI is centered and the aircraft is directly over the airborne checkpoint?

- ☐ Plus 6degrees or minus 4 degrees of the designated radial.
- ☐ Plus or minus 6 degrees of the designated radial.
- ☒ Plus or minus 4 degrees of the designated radial.

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24

Which may lead to a power push-over in a gyroplane?

- ☐ Low speed.
- ☐ Decreasing power too quickly.
- ☒ Rotor force is removed.

25

Identify runway remaining lighting on centerline lighting systems.

- ☐ Amber lights from 3,000 feet to 1,000 feet, then alternate red and white lights to the end.
- ☐ Alternate red and white lights from 3,000 feet to the end of the runway.
- ☒ Alternate red and white lights from 3,000 feet to 1,000 feet, then red lights to the end.

26

A possible result of using the emergency alternate source of static pressure inside the cabin of an unpressurized airplane is the

- ☒ altimeter may indicate an altitude lower than the actual altitude being flown.
- ☐ airspeed indicator may indicate less than normal.
- ☐ altimeter may indicate an altitude higher than the actual altitude being flown.

27

The disadvantages of a float type fuel gauge are:

- 1- the design is complex
- 2- the indications are influenced by the aircraft attitude variations
- 3- the indications are influenced by the accelerations
- 4- the indications are influenced by temperature variations

The combination that regroups all of the correct statements is:

☐ 1, 2, 3, 4.

☒ 2, 3.

☐ 1.

☐ 2, 3, 4.

28

An aircraft is flying from a cold air mass into a warm air mass. The TAS and true altitude will:

☐ TAS decreases, true altitude increases.

☒ Both increases.

☐ TAS increases, true altitude decreases.

29

If the static vent became blocked during a descent, the altimeter would:

☐ under-read the correct altitude

☐ over-read the correct altitude

☒ act normally

☐ read zero altitude

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30

The high Altitude Radio Altimeter uses the following wavelengths:

☐ **decimetric.**

☐ **myriametric.**

☐ **hectometric.**

☒ **metric.**

31

The value of the compressibility error of an Airspeed Indicator (ASI) will:

- ☒ **decrease, with decrease in density and increase with an increase in IAS;**
- ☐ **increase with increase in altitude and true airspeed;**
- ☐ **increase with increase in true airspeed and decrease with increase in altitude.**

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32

Rectified Airspeed is:

- ☐ IAS corrected for density error.
- ☐ IAS corrected for instrument and pressure error.
- ☒ IAS corrected for density and compressibility errors.

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33

The limits of the green scale of an airspeed indicator are:

☐ *VS1 and VMO.*

☐ *VS1 and VNE.*

☐ *VS0 and VNE.*

☒ *VS1 and VNO.*

34

A machmeter does not suffer from:

- ☐ instrument error
- ☐ error due to blockage of the static vent
- ☒ compressibility error
- ☒ position error

35

In a CADC the output of vertical speed is achieved by means of:

- ☒ **a measurement of the rate of change of altitude as sensed by the static pressure transducer**
- ☐ **an integration of the CAS transducer output**
- ☐ **a dedicated vertical speed transducer**
- ☒ **a combination of the outputs of the dynamic and static pressure transducers**

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36

The magnetic system of a compass is suspended pendulously to counteract the effect of component:

☒ Z

☐ P

☒ H

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37

The deviations of a direct reading magnetic compass effected only by coefficients B and C are, 3 E on 230 (C) and 4 W on 130 (C), therefore the deviation on heading 315 (C) is:

☐ 7,5 W

☐ 3,3 W

☒ 3,3 E

38

The effect of decreasing rotor speed in the turn and slip indicator will cause:

- ☒ the turn indicator to under-read the rate of turn.
- ☐ the turn indicator to under-indicate the angle of bank but will not effect the slip indicator;
- ☒ the turn indicator to over-read and the slip indicator to under-read;

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39

During a stabilised climbing turn at a constant rate, the instruments which indicate the correct pitch and bank are the:

☒ vertical-speed indicator and turn-and-slip indicator;

☐ attitude indicator and turn-and-slip indicator.

☐ altimeter and turn-and-slip indicator;

40

The parameters taken into account by the flight director computer in the altitude hold mode (ALT HOLD) are:

- 1 - altitude deviation
- 2 - engine rpm
- 3 - ground speed
- 4 - pitch attitude

The combination that regroups all of the correct statements is:

- ☐ 1.
- ☐ 1, 2.
- ☒ 1, 4.
- ☐ 1, 3.

